

Prime Factorization

Flagship

Lesson 1-1-flagship

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



factors: 1 and 7

*7 has only two factors:  $1 \times 7$ . So 7 is prime.*

### Prime number

Write the definition:

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$6 = 1, 2, 3, 6$

many factors  $\rightarrow$  composite

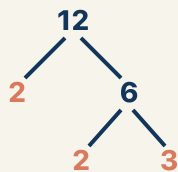
*$12 = 1 \times 12, 2 \times 6, 3 \times 4$  — six factors, so 12 is composite*

### Composite number

Write the definition:

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$$12 = 2 \times 2 \times 3$$

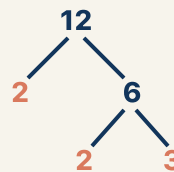
$$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$

### Prime factorization

Write the definition:

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$$12 = 2 \times 2 \times 3$$

$$24 \rightarrow 4 \times 6 \rightarrow (2 \times 2) \times (2 \times 3) \rightarrow 2 \times 2 \times 2 \times 3$$

### Factor tree

Write the definition:

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$$5^3$$

$$5 \times 5 \times 5$$

$2^3$  means  $2 \times 2 \times 2 = 8$

## Exponent

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

I can write a number as a product of its prime factors using a factor tree.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Prime number

Composite number

Prime factorization

Factor tree

Exponent

1

A whole number greater than 1 with exactly two factors, 1 and itself, is a \_\_\_\_\_ number.

2

A whole number greater than 1 that has more than two factors is a \_\_\_\_\_ number.

3

\_\_\_\_\_ — Writing a number as prime numbers multiplied together.

4

I can break a number into its prime factors step by step using a \_\_\_\_\_.

5

In  $2^3$ , the small number 3 is the \_\_\_\_\_ and it shows 2 is multiplied 3 times.



### Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** Which of the following is a prime number?

- A. 17
- B. 15
- C. 21
- D. 9

**Step 1** 17 has exactly two factors: 1 and 17.

**Step 2**  $15 = 3 \times 5$ ,  $21 = 3 \times 7$ , and  $9 = 3 \times 3$ , so they are all composite.

 **Answer:** A. 17

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** What is the prime factorization of 30?

- A.  $2 \times 3 \times 5$
- B.  $5 \times 6$
- C.  $2 \times 15$
- D.  $3 \times 10$

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** What is the prime factorization of 18?

- A.  $2 \times 3 \times 3$
- B.  $2 \times 9$
- C.  $3 \times 6$
- D.  $6 \times 3$

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Lock 3 — Rewrite the prime factorization of 72 in exponent form.**

A.  $2^3 \times 3^2$

B.  $2^2 \times 3^3$

C.  $2 \times 3^5$

D.  $6^2$

Show your work:

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**2. Final Vault — The vault code is the prime factorization of 84 with all prime factors listed. Which code opens it?**

A.  $2 \times 2 \times 3 \times 7$

B.  $2 \times 42$

C.  $4 \times 21$

D.  $2 \times 2 \times 21$

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.**

*Sentence starter: I chose the number \_\_\_\_\_. My first factor tree starts with \_\_\_\_\_  $\times$  \_\_\_\_\_, and my second starts with \_\_\_\_\_  $\times$  \_\_\_\_\_. Both give the same prime factorization: \_\_\_\_\_. This happens because \_\_\_\_\_.*

Show your work:

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## Reflect — Exit Ticket

**What is the prime factorization of 40?**

- A.  $2 \times 2 \times 2 \times 5$
- B.  $4 \times 10$
- C.  $5 \times 8$
- D.  $2 \times 20$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor tree
5. **Notes 5:** Exponent
6. **We Try (notes):** A.  $2 \times 3 \times 5 = 30 = 2 \times 15 = 2 \times 3 \times 5$ . All three factors (2, 3, 5) are prime, so  $2 \times 3 \times 5$  is the prime factorization.
7. **You Try (notes):** A.  $2 \times 3 \times 3 = 18 = 2 \times 9 = 2 \times 3 \times 3$ . Both 2 and 3 are prime, so  $2 \times 3 \times 3$  is the prime factorization.
8. **Try It 1:** A.  $2^3 \times 3^2 = 72 = 2 \times 2 \times 2 \times 3 \times 3$ . Three 2s and two 3s give  $2^3 \times 3^2$ .
9. **Try It 2:** A.  $2 \times 2 \times 3 \times 7 = 84 = 4 \times 21 = (2 \times 2) \times (3 \times 7) = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$ . The vault opens!
10. **Exit Ticket:** A.  $2 \times 2 \times 2 \times 5 = 40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$ . All factors (2, 2, 2, 5) are prime.